enough to touch, and at that same first moment, rapidly tearing away from one another.

The logic of the Big Bang theory is pretty simple. The universe is expanding—we can see that distances between galaxies are getting larger over time—which means that the distances between galaxies were smaller in the past. We can, as a thought experiment, rewind the expansion we see now, extrapolating back across billions of years, until we reach a moment when the distance between galaxies must have been zero. The observable universe, encompassing everything we can see today, must have been contained within a much smaller, denser, hotter space. But the observable universe is just the part of the cosmos we can see now. We know that space goes on much farther than that. In fact, based on what we know, it's entirely possible, and perhaps probable, that the universe is infinite in size. Which means that it was infinite at the beginning too. Just much denser.

This is not easy to picture. Infinities are tough that way. What does it mean to have infinite space? What does it mean for an infinite space to be expanding? How does infinite space get infinit*er*?

I'm afraid I can't help you with this.

There is simply no easy way to hold infinite space in a finite brain. What I can say is that there are ways to deal with infinities in mathematics and physics that make sense and don't break anything. As a cosmologist, I work from the basic assumption that the universe can be described with math, and if that math works out, and is useful for approaching new problems, I go with it.[fn4] Or, more precisely, if the math works out and a somewhat different assumption (e.g., that the universe is not *quite* infinite but is so big that we can't possibly ever perceive its edges) also works but makes no difference to our experience or anything we can measure in any way, we may as well stick with the simpler assumption for now. So: infinite universe. We can work with that.

In any case, when we talk about the Big Bang theory, what we're really saying is: based on our observations of the present expansion and its history, we can conclude that there was a time when the universe was, everywhere, much hotter and denser than it is today.[fn5] This is sometimes called the "Hot Big Bang," referring to the whole span of time when the universe was hot and dense, which we now know to be the time from year 0 to somewhere around year 380,000.[fn6]